

Cloud Computing & DevOps – AWS Handbook

SESSION 10

Kubernetes Fundamentals

Container Orchestration and Cluster Management

Duration	2 Hours
Prerequisites	Docker Fundamentals and Docker Compose
Learning Outcome	Understand Kubernetes architecture, deploy applications, manage pods, deployments and services.

Why Are We Learning This?

As applications grow, managing containers manually becomes difficult. Kubernetes automates deployment, scaling, load balancing, recovery, and management of containerized applications across clusters of servers.

Architecture Diagram



Key Concepts

Cluster	Collection of nodes managed by Kubernetes.
Node	Machine that runs containers.
Pod	Smallest deployable Kubernetes unit.
Deployment	Manages pod lifecycle and scaling.
Service	Provides network access to pods.
Control Plane	Manages cluster operations.

Practical Activities

- Explore Kubernetes cluster architecture
- Deploy a pod
- Create a deployment
- Scale the deployment
- Create a service
- Access application through the service
- Verify running resources

Expected Outputs

Pods deployed successfully, deployment created, service accessible, and application available through Kubernetes networking.

What You Should See

Paste Screenshot Here

Cluster Information

Paste Screenshot Here

Pod Deployment Output

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Deployment Details

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Service Configuration

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Application Running via Kubernetes

Journal Writing Template

Aim

To deploy and manage applications using Kubernetes.

Theory

Kubernetes is a container orchestration platform that automates deployment, scaling and management of containers.

Procedure

Write 5–7 concise steps performed during the practical session.

Observations

Pods were deployed successfully, services provided connectivity and deployments enabled application management.

Conclusion

Kubernetes simplified deployment and management of containerized applications.

Viva Preparation

1. What is Kubernetes?
2. What is a Pod?
3. What is a Deployment?
4. What is a Service?
5. What is a Cluster?
6. What is a Node?
7. Why is Kubernetes required when Docker already exists?

Troubleshooting Box

Problem	Possible Cause
Pod not running	Image or configuration issue
Service inaccessible	Networking issue
Deployment failed	Manifest configuration error
Application unavailable	Container startup failure

Industry Relevance

Kubernetes is the industry standard platform for container orchestration and is used by organizations worldwide to run scalable cloud-native applications.

Session Summary

Students learned Kubernetes fundamentals, including pods, deployments, services, and the basics of container orchestration.